STACKCALC SC-32 SCIENTIFIC RPN CALCULATOR

4-Register RPN Stack Engine, Dual Stand Modes & Hardware Guide

STUDIO CONTINUUM: 1·Ten-Frames | 2·Dominoes | 3·Clock | 4·Fractions | 5·Primes | 6·Algebra | 7·Probability | 8·Volvelle | 9·Quest | 10·Rack | 11·SC-32

CARD 09 / 10 • SC-32-HW

MATHEMATICAL REFERENCE

1. HARDWARE ARCHITECTURE & STANDS

4-REGISTER RPN ENGINE & DUAL STANDS

Reverse Polish Notation (RPN) with 4 registers.
Numbers push onto stack; operators evaluate at once:
Stack Architecture: Registers T, Z, Y, X
Zero parentheses, zero ambiguity, zero errors.

CALIBRATED 32-KEYCAP TACTILE HARDWARE

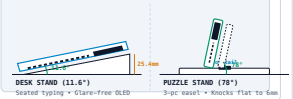
- Display: 128×64 OLED with live 4-stack view
- Keycaps: 43 sculpted Shore 90A TPU keys over tactile switches
- Stack: Push on [ENTER], pop on op, LastX saved
- Power: USB-C LiPo with <15 µA ultra-low sleep

DUAL-MODE DESK & PUZZLE STANDS

- Desk Stand: 11.6° forward incline for typing.
- Puzzle Stand: 3-piece easel docks unit at 78°.

Knocks flat to 6mm to slip into travel caddy.

OPERATING MODES: 11.6° DESK INCLINE & 78° PUZZLE EASEL



DESK STAND (11.6°)
Seated typing • Glare-free OLED

PUZZLE STAND (78°)
3-pc easel • Knocks flat to 6mm

2. KIT PARTS INVENTORY (6 PIECES)

COMPONENT	QTY	SPEC / DIMENSION	PALETTE COLOR
SC-32 Scientific Unit	1x	128×64 OLED, 32 Keycaps	Matte Obsidian
TPU Protective Sleeve	1x	Shore 95A Bumper Cushion	Flexible Black TPU
Interlocking Puzzle Stand	1x	3-Piece 78° Portrait Easel	Matte Charcoal
Desk Stand Caddy	1x	11.6° Incline Operating Dock	Matte Charcoal
USB-C Charging Cable	1x	Braided 1.0m Power Cable	Stealth Black
Mailer Storage Box	1x	6×4 Inch Protective Case	Kraft Cardboard

TOTAL: Calculator + TPU Sleeve + Puzzle Stand + Desk Caddy + Cable + Box = 6 Parts

POWER CONTROL & ANNUNCIATORS

- Instant Power Off: Tap [f] then [ON/OFF] to shut down.
- Auto-Sleep Standby: Sleeps after 5 min idle (<15 µA).
- Non-Volatile Memory: Registers (X, Y, Z, T) persist.
- Annunciators: RAD/DEG, [f], [g], battery on OLED header.

3. WHY KEEP THIS REFERENCE CARD?

✓ **REPLACEMENT STAND & HARDWARE**

Need replacement puzzle stand trusses, a fresh TPU sleeve, keycaps, or braided USB-C cable? All physical hardware components are stocked.

Scan back QR or visit stackcalc.io/parts
Order individual replacement pieces anytime

✓ **NO LOOSE BLOCKS — JUST TURN IT OFF!**

Unlike manipulative block kits, you don't need to count or pack up 70 pieces! Simply press [f] [OFF] to turn off, or let auto-sleep preserve your stack. Dock in the Desk Stand or slip into the TPU sleeve and drop in your backpack. Your calculation state waits in memory until your next keystroke!

✓ **HANDS-ON TUTORIAL ON REVERSE**

Flip this card over for a complete step-by-step hands-on lesson with concrete numbers, stage arrangements, and calculator cross-references!

Keep this card stored inside the box lid!

STACKCALC ARCHIVAL INSTRUMENT LABS

Designed in Mountain View, California, USA • Non-Toxic BPA-Free.
Classroom Lab Crates (10x / 30x Stations): stackcalc.io/schools

POSTFIX EXPRESSION EQUIVALENCE THEOREM

Any algebraic expression can be evaluated unambiguously without parentheses using a postfix stack.

Universal 7.00mm Datum • Classroom Continuum (10 Kits)

OFFICIAL REFILL PORTAL

stackcalc.io/parts
Order replacement parts anytime

SERIES 1 • CARD 09 OF 10 • DOMAIN: 4-REGISTER RPN COMPUTING & EXPRESSION TREES • HARDWARE TACTILE KEYS • DUAL LED BAR GRAPH STACK

LESSON GUIDE: 4-REGISTER RPN STACK & EXPRESSION TREES

Step-by-Step Hands-On Tutorial: Pythagorean Hypotenuse $\sqrt{(3^2 + 4^2)} = 5$

Grades 8–16 STEM, Physics & Engineering • Hands-On Classroom & Home STEM Studio Series

CARD 09 / 10 • SC-32-HW

MATHEMATICAL GUIDE

SCAN FOR 3D VIDEO DEMO

OPEN CAMERA & SCAN
stackcalc.io/products/sc32-calculator/
Zero app install • iPad & Chromebook

CURRICULAR ALIGNMENT

- CCSS.MATH.PRACTICE.MP5 (Use Tools Strategically)
- CCSS.MATH.PRACTICE.MP6 (Attend to Precision)
- TEKS 111.39.c.1.C (Manipulatives & Technology)

Grades 8–16 STEM, Physics & Engineering

MATHEMATICAL INQUIRY

Where does the first number wait safely on the stack while you are entering the second number?

1 STEP 1: ENTER 3 AND 4 ONTO STACK

Power on the SC-32. Key: 3 [ENTER] 4 [ENTER].
Stack state: Register Y holds 3.0000; Register X holds 4.0000.
In RPN, numbers always enter before operators!

2 STEP 2: SQUARE BOTH REGISTERS

Press [↷] [x²] on 4: Register X becomes 16.0000.
Press swap [⇄y]: Register X becomes 3.0000; Y holds 16.0000.
Press [↷] [x²] on 3: Register X becomes 9.0000.
Stack now holds Y = 16.0000 and X = 9.0000.

3-4 STEPS 3 & 4: COLLAPSE STACK & SQRT

3. ADD SQUARES: Press [+]. The stack drops: 16.0000 + 9.0000 = 25.0000 in X!
4. SQUARE ROOT: Press [√x]. Result: 5.0000!
Combined Keystrokes: 3 [↷] [x²] 4 [↷] [x²] [+] [√x] → 5.

5 CORE PRINCIPLE: THE FREEDOM FROM PARENTHESES

Evaluating $\sqrt{(3^2 + 4^2)}$ in traditional algebraic mode requires nested parentheses. In RPN, you calculate from the inside out, exactly as you think! Intermediate numbers sit patiently on the stack until an operator consumes them. Zero parentheses, zero syntax errors!

★ EXTENSION INVESTIGATION: KINETIC ENERGY: $E = \frac{1}{2} m v^2$

Calculate kinetic energy for mass m=12 and velocity v=5: 0.5 [ENTER] 12 [×] 5 [↷] [x²] [×] →
150.0000! Zero parentheses! (Scan QR for 3D animated solution!)
→ Assemble with your physical blocks, then scan the QR code to verify with the 3D step-by-step vector model.

STACKCALC ARCHIVAL INSTRUMENT LABS

Interactive Mathematics Manipulative Ecosystem.
Designed in Mountain View, California • Archival BPA-Free POM/PETG.

CROSS-REFERENCED CALCULATOR VERIFICATION

Every keystroke sequence verified with RPNCore CalculatorEngine.
Modular Hardware Continuum (9 Kits) • Universal 7.00mm Pitch • stackcalc.io

PARTS & SCHOOL REFILL PORTAL

stackcalc.io/parts
Classroom Lab Crates: stackcalc.io/schools

STACKCALC SC-32 HARDWARE, POWER & DUAL STANDS

DESK INCLINE (11.6°) • PUZZLE EASEL (78°)

1. POWER & AUTO-SLEEP

INSTANT SHUTDOWN: [f] [OFF]
Electronic calculator with zero loose blocks to pack. Tap [f] then [ON/OFF] to shut down immediately.

ULTRA-LOW AUTO-SLEEP
Standby engages automatically after 5 minutes idle (<15 µA draw). Battery lasts weeks on standby.

NON-VOLATILE FLASH RAM
Registers (X, Y, Z, T), LastX, angular modes, and memories persist completely across power cycles.

INSTANT RESUME ON KEY
Press [ON] or tap any key to wake immediately with previous calculation waiting where you left it.

ZERO PACK-UP OVERHEAD
Turn it off, slip into TPU sleeve, and go.

2. DESK STAND (11.6°)

1.0 IN REAR ELEVATION
Elevates rear by 25.4 mm, giving an optimal 11.6° forward slope for seated one-handed typing.

ANTI-GLARE OLED VIEW
Angled viewing geometry eliminates ceiling fluorescent light reflections across the OLED panel.

INTEGRATED CABLE TROUGH
Rear pass-through connects braided USB-C cable for continuous desktop charging during operation.

ROCK-SOLID FOOTPRINT
Wide stance chassis with non-slip base prevents sliding or tipping during rapid tactile key entries.

ACTIVE OPERATION MODE
Keep docked on desk alongside your workspace.

3. PUZZLE STAND (78°)

3-PIECE FRICTION DOVETAIL
Tool-free interlocking trusses snap together by hand in 5 seconds with precision 0.20 mm slide tolerance.

UPRIGHT PORTRAIT EASEL
Docks calculator at 78° upright portrait viewing beside open STEM textbooks and lab worksheets.

DUAL INTEGRATED BAYS
Front cradle holds calculator securely; rear bay docks TPU travel sleeve or reference card upright.

KNOCKS FLAT TO 6.0 MM
Disassembles in 2 seconds into 3 slim plates that slip flat into a pencil case or backpack pouch.

LAB & SHELF DISPLAY
Optimal reading angle while hands are full.

HARDWARE WORKFLOW & TRAVEL SETUP

1. SEATED DESK CALCULATING (DESK STAND):
Set travel caddy on desk and seat calculator onto 11.6° incline. Connect USB-C for continuous power during long study sessions.

2. HANDS-ON LAB INVESTIGATIONS (PUZZLE STAND):
Slide the two puzzle stand trusses into cradle shelf until flush. Dock calculator at 78° to read values while building physical models.

3. POWER DOWN & SLEEP:
When finished, press [f] [OFF] to shut down, or let auto-sleep preserve your stack. There are no loose parts to count or pack!

4. TRAVEL PROTECTION & KNOCKDOWN:
Slip calculator into Shore 95A TPU bumper. Knock puzzle stand flat to 6mm. Pack both into travel caddy for drop-proof transit.

SC-32-HW: HARDWARE SPECIFICATION & QC AUDIT

CALIFORNIA LAB QC

CALIFORNIA LAB QC HARDWARE AUDIT

[✓] OLED DISPLAY & POLARIZER AUDIT:
128×64 high-contrast display verified for zero pixel defects, uniform brightness, and 4-line stack viewing.

[✓] 43 TACTILE SWITCHES & TPU KEYS:
Actuation force verified at 1.2 N ± 0.1 N with positive tactile feedback across all 43 keys.

[✓] PUZZLE STAND DOVETAIL SLIDE FIT:
Interlocking trusses seat flush with 0.20 mm precision friction fit for wobble-free 78° docking.

[✓] USB-C POWER DELIVERY & SLEEP CURRENT:
LiPo charging controller and ultra-low standby current (<15 µA) verified across power-down cycles.

[✓] SHORE 95A FLEXIBLE TPU MEMBRANE:
Perimeter shock-absorbing bumper cushion molded to exact 0.25 mm clearance for drop protection.

[✓] STACK STATE MEMORY RETENTION:
Registers X, Y, Z, T, LastX, and angular modes tested for 100% data integrity through simulated auto-sleep.

RPN HARDWARE COMPATIBILITY

- Full scientific transcendental engine (sin, cos, tan, ln, e^x, y^x)
- Statistical registers ($\Sigma+$, $\Sigma-$, mean, standard deviation)
- Postfix stack eliminates operator precedence ambiguity
- Digital twin of physical manipulative heights across all kits!

TOTAL HARDWARE INVENTORY: 6 PIECES

Verified by Electronics Tech & Hardware Test Bench.

Inspection Pass • Mountain View, CA

LIFETIME HARDWARE GUARANTEE

NO COMPONENT LEFT BEHIND™

Hardware components wear or get misplaced. We've got you covered. Replacement puzzle stand trusses (piece_stand_front, piece_stand_rear), TPU protective sleeves, and braided USB-C cables are permanently stocked at stackcalc.io/parts with same-day dispatch ($1.99–$7.50).
PORTAL: stackcalc.io/parts

DISTRICT & CLASSROOM PROCUREMENT

- Net-30 Invoicing for US Public & Private School Districts.
- Volume Classroom Bundles (10-packs & 30-packs).
- Free Teacher Activity Books & Lesson Blueprints.
- Title I STEM & Carl D. Perkins Grant eligible.

Email: procurement@stackcalc.io
stackcalc.io/schools

3D HARDWARE INTERACTIVE DEMOS

Scan the QR code on the back of this card (Side B) to access:

- 60fps 3D animated walkthrough of RPN stack dynamics
- Interactive 3D explode view of SC-32 chassis & stand
- Step-by-step calculator keystroke simulations
- Printable student worksheets and answer keys

VIDEOS: stackcalc.io/products

Designed & Engineered in Mountain View, California